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Business Analytics

Experiment No.: 2

Title: Portfolio Profitability Architecture and Multi-Page Executive Performance Reporting inside Power BI Desktop

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Experiment No.: 1

Portfolio Profitability Architecture and Multi-Page Executive Performance Reporting inside Power BI Desktop

Problem Statement

An enterprise retail organization requires a comprehensive multi-dimensional analytical platform to dynamically monitor, evaluate, and interpret commercial sales performance data. The corporate governance framework necessitates a highly granular visualization layer that cross-examines key performance metrics across vast product portfolios, structural sub-categories, and distributed geographic operational nodes.

As a Lead Data Analyst, your objective is to connect Power BI Desktop to the historical transactional database engine, perform extract-transform-load (ETL) processing routines inside Power Query, deploy safe functional DAX computational metrics, and design a fully synchronized 2-page executive report. The system must isolate top line margins while explicitly mapping profit deficit segments and temporal consumer seasonality.

Methodology / Steps Followed

Step 1: Data Collection & Storage Schema Ingestion

To model a robust analytical environment, the global **Sample Superstore** relational database ledger was utilized. The file architecture captures full transactional fields including order keys, timeline logs, customer profile nodes, categorical groupings, and financial variables (Sales, Quantity, Discount, and Profit).

- Pipeline Initialization: Launched the Power BI Desktop data connection interface via the *Get Data* pipeline.
- Ingestion Configuration: Selected the local file network connector to point to the raw master table sheet.
- Storage Mode Optimization: The data model was explicitly loaded utilizing the **Import Connectivity Mode**. This choice guarantees advanced column-store compression, loads data indexes into the highly reactive local VertiPaQ caching engine, and ensures lightning-fast filter rendering across charts.

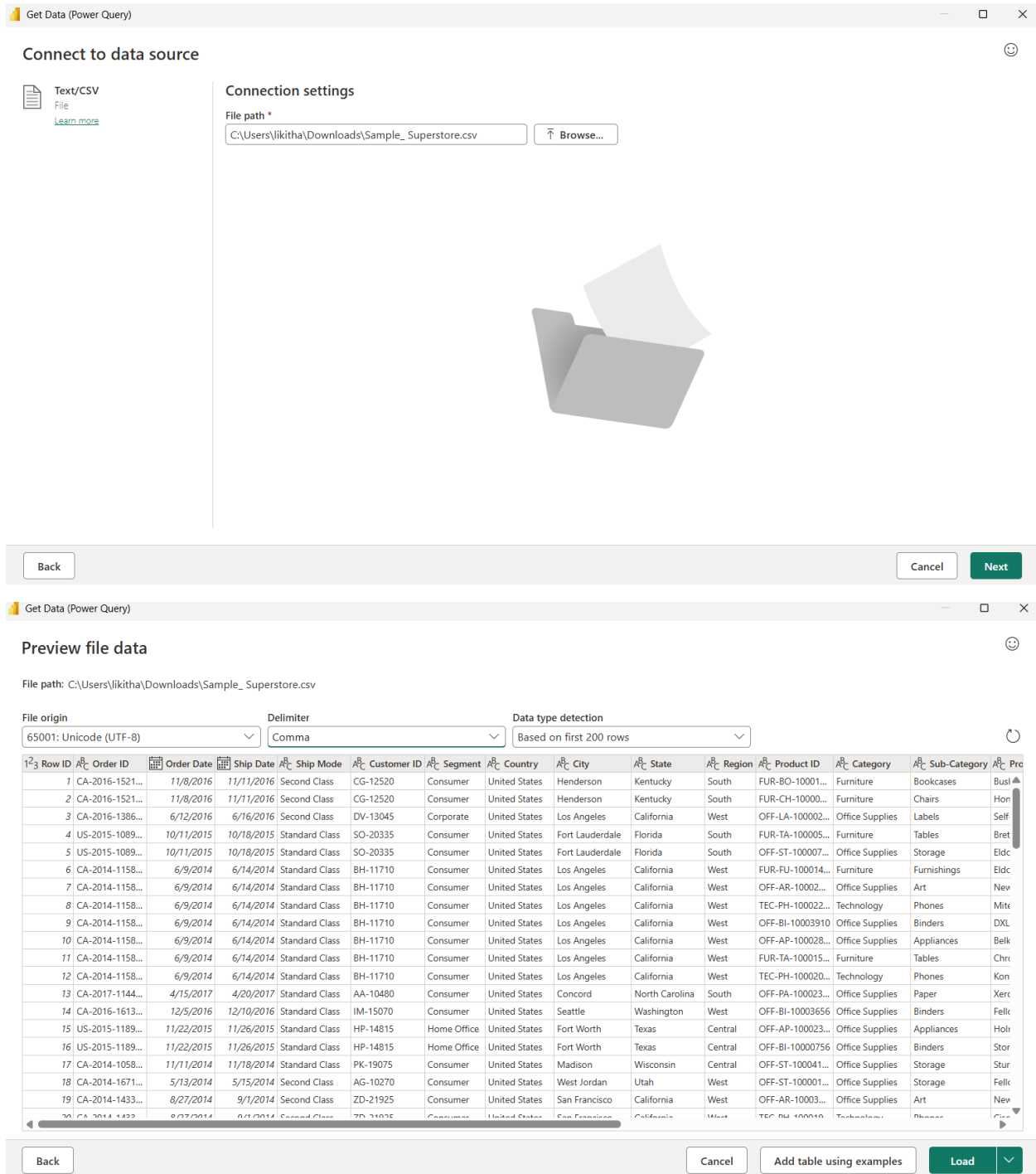
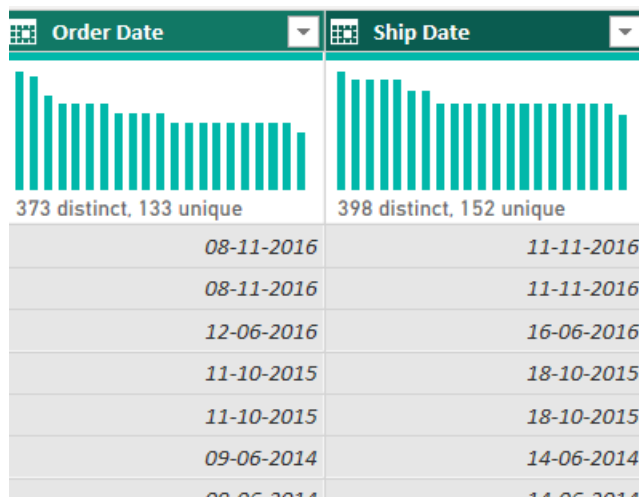


Figure 1: Ingestion Module & Storage Connection Configuration Screen

Step 2: Advanced ETL Data Cleaning and Time Engineering

Upon loading the relational array, the raw dataset was migrated directly into the **Power Query Editor** to enforce strict semantic rules and prevent downstream data type exceptions:

- **Attribute Rectification:** The column data type for the Postal Code field was manually adjusted from a standard whole number index to a strict **Text string** configuration. This prevents the modeling engine from executing math aggregations on zip codes and preserves mandatory leading zero structures.
- **Numeric Verification:** Verified that monetary transaction scales (Sales and Profit) were structurally encoded as **Decimal Numbers**, while unit counts (Quantity) were fixed as **Whole Numbers** to eliminate processing truncation.
- **Temporal Feature Engineering:** To accommodate multi-line comparative line chart visualization over rolling cycles, granular variables were engineered. Selecting the baseline Order Date column, the *Add Column* date menu was utilized to extract two standalone chronological attributes: Year and Month Name (structured to sort sequentially from January to December).



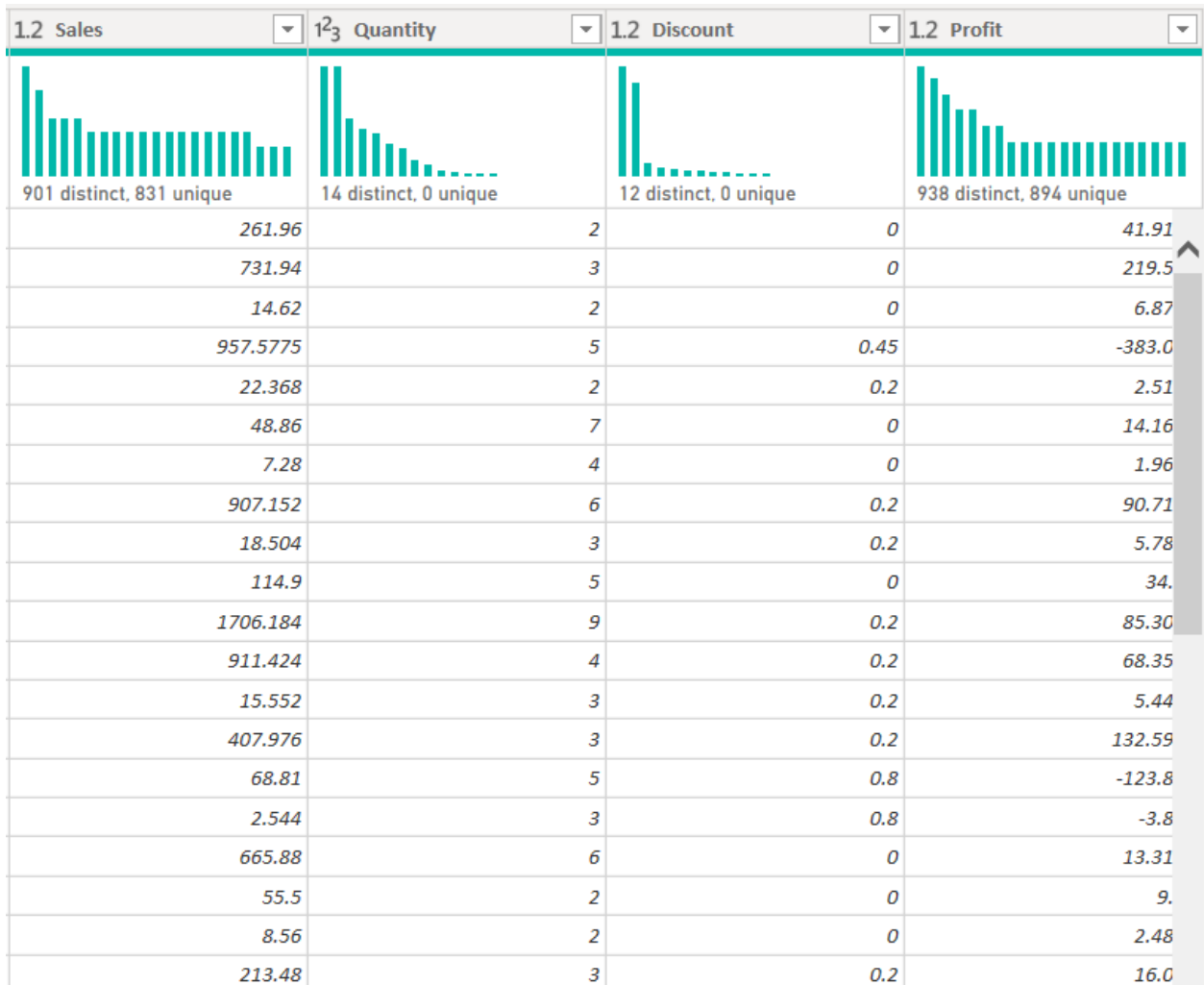


Figure 2: Power Query Data Processing and Applied Transformation Steps Panel

Step 3: Analytical DAX Model Engineering

To prevent report visualization crashes from division-by-zero or empty-set errors within large datasets, a dedicated calculated measures repository named `_Measures` was constructed. The underlying business logic formulas were written utilizing safe functional context routing:

1. Total Sales Pipeline (Gross Revenue Variable):
Total Sales = SUM('Sample Superstore'[Sales])
2. Net Organizational Profitability Ledger:
Total Profit = SUM('Sample Superstore'[Profit])

3. Net Profit Margin Percentage (Zero-Safe Logic):
Profit Margin % = DIVIDE([Total Profit], [Total Sales], 0)

(The DIVIDE function dynamically handles cases with null values or zero denominators by substituting an alternative blank result instead of returning an infinity calculation error.)

Step 4: Dashboard Development

An advanced, multi-page analytical viewport interface was engineered within the Report View canvas to establish a synchronized financial reporting environment:

- Strategic KPI Scorecard Grid: Placed three distinct card visuals at the top left mapping total global sales metrics (2.30M), corporate net profit margins (286.40K), and the safe margin efficiency ratio (12.47%).
- Temporal and Demographics Control Section: Embedded an interactive Between date timeline slider alongside a compact Dropdown selection filter for Region in the upper-right quadrant to drive instantaneous cross-filtering across the entire page.
- Categorical Market Share Analytics: Deployed a polished Donut chart showcasing total sales distribution by main categories (Technology, Furniture, Office Supplies) featuring explicit percentage metrics labels.
- Geographic Centers: Structured a horizontal Clustered Bar chart utilizing **Top N filter rules** inside the metadata pane to limit display rows to the Top 10 revenue-generating states.
- Chronological Volume Trends: Positioned a multi-year Line Chart stretching across the lower layout area, overlapping performance tracks across annual boundaries to evaluate operational velocity.
- Customer Segment Button Panel: Structured an interactive horizontal slicing ribbon on Page 2 using a **Tile** display format for immediate selection of Consumer, Corporate, and Home Office filters.
- Portfolio Loss Isolation Graph: Configured a left-aligned Clustered Bar chart set to **Ascending sort order** based on the Total Profit measure to force the highest deficit-producing items straight to the top of the visibility hierarchy.
- Granular Financial Portfolio Matrix: Positioned a deep-dive multi-layered Matrix table on the right side of the canvas. The row layout embeds a collapsible parent-child hierarchy (Category > Sub-Category) supporting deep drill-downs, while the values block displays Total Sales, Total Profit, and Profit Margin % mapped with custom conditional red-to-green color scales.

Output

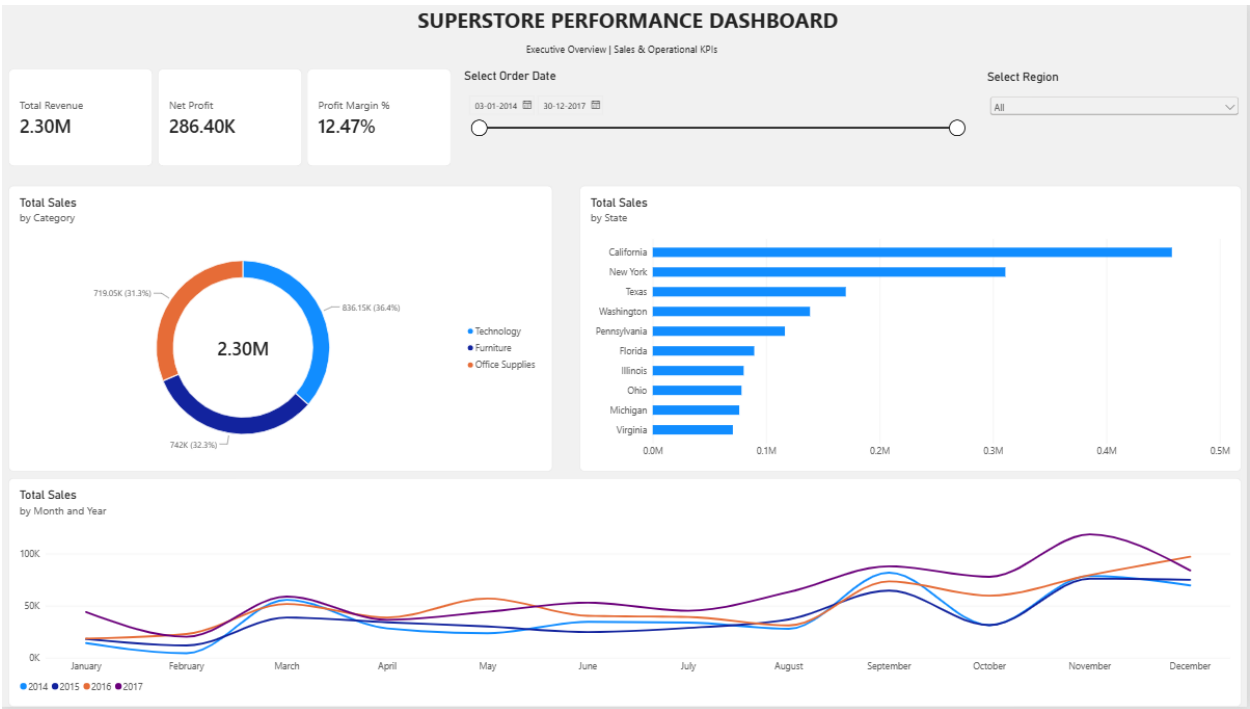


Figure 3: Completed Executive Overview Reporting Layout (Page 1)

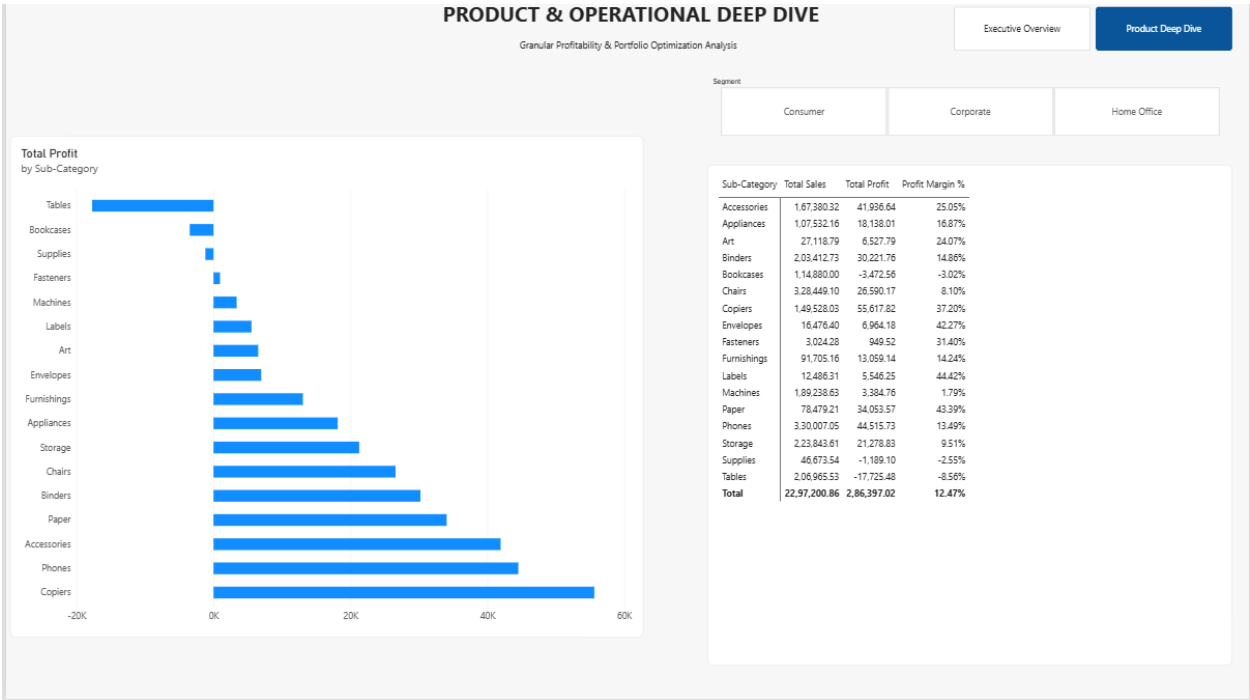


Figure 4: Completed Product Portfolio & Profit Diagnostic Matrix Layout (Page 2)

The completed enterprise Power BI dashboard dynamically renders comprehensive commercial retail metrics:

- Total Sales Performance: A comprehensive \$2.30M gross transaction value processed across the multi-year business operation window.
- Customer Segment Analysis: Advanced cross-filtering allows dynamic drill-down across specific Consumer, Corporate, and Home Office pipelines.
- Product-wise Sales and Profit Analytics: Highlights portfolio performance while isolating heavy revenue engines from low-performing deficit anchors.
- Regional Sales Performance: Tracks spatial distribution across major domestic geographic centers.
- Monthly Sales Trends: Maps multi-year linear transaction timelines to check demand velocity.
- Interactive Filters and Slicers: Synchronized global controls recalculate report visual panels dynamically.

The dashboard enables management to monitor business performance and make data-driven decisions effectively.

Business Insights

Deep evaluation of the visual data metrics uncovers strategic structural opportunities:

- Geographic Sales Concentrations: Enterprise revenue is heavily anchored within the West and East territories. Deep filtering confirms that **California** and **New York** serve as the two primary commercial hubs driving corporate revenue pipelines.
- Structural Portfolio Risk (Profit Leaks): Granular analysis reveals a significant financial vulnerability within the Furniture division. While generating healthy sales volumes, the **Tables sub-category** operates at a massive net loss. This profit leak is driven by aggressive discount rates that compress net revenue past sustainable operational cost margins.
- Temporal Demand Seasonality: Multi-year chronological linear trend comparisons reveal sharp, recurring transaction volume spikes concentrated heavily throughout **Q4 (specifically November and December)**. Supply chain operations should systematically align inventory ramp-ups and shipping allocations ahead of this holiday wave.

Conclusion

This laboratory experiment successfully confirms the execution of an end-to-end data processing workflow. By pulling messy transactional flat data into an optimized, multi-page data app, the final dashboard turns complex raw rows into actionable, real-time metrics. The system provides executive leadership with clear visibility to optimize corporate pricing rules, eliminate product portfolio deficits, and maximize revenue streams over seasonal cycles.

Learning Outcomes

Through this project, I learned:

- How to configure high-efficiency data asset models under **Import Storage Mode** rules.
- How to perform chronological data cleaning and feature engineering transformations using Power Query.
- How to establish core star-schema mapping strategies to handle structured transactional models.
- How to design safe, error-free calculated fields and metric logic using advanced DAX language components.
- How to manage multi-page dashboard layouts using buttons and dynamic navigation elements.
- How to translate visual data matrices into concrete, actionable business recommendations for executive leadership.